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important
kubernetes
cmds



kubectl get

list resources
(pods, services, nodes, deployments)

```
user@k8s:$ kubectl get pods           # list all pods
user@k8s:$ kubectl get services        # list all services
user@k8s:$ kubectl get deployments    # list all deployments
user@k8s:$ kubectl get nodes           # list all nodes
user@k8s:$ kubectl get pods -o wide -A # all pods in all namespaces
```



kubectl describe

show detailed information
about a specific resource

```
user@k8s:$ kubectl describe pod <pod-name>
```

```
# details of a specific pod in default namespace
```

```
user@k8s:$ kubectl describe service <service-name>
```

```
# details of a specific service in default namespace
```

```
user@k8s:$ kubectl describe pod <pod-name> -n <namespace>
```

```
# details of a specific pod in specific namespace
```



kubectl apply

create or update resources
using a YAML file

```
user@k8s:$ kubectl apply -f <file.yaml>  
# apply changes from a YAML file
```



kubectl delete

remove resources
from the cluster

```
user@k8s:$ kubectl delete pod <pod-name>
```

```
# delete a specific pod
```

```
user@k8s:$ kubectl delete -f <file.yaml>
```

```
# delete resources defined in a YAML file
```



kubectl logs

view logs from
a specific pod or container

```
user@k8s:$ kubectl logs <pod-name>
```

```
# logs from a single-container pod
```

```
user@k8s:$ kubectl logs <pod-name> -c <container-name>
```

```
# logs from a specific container in a pod
```



kubectl exec

execute a command
inside a running pod

```
user@k8s:$ kubectl exec -it <pod-name> -- /bin/bash
```

```
# start a bash shell in a pod
```

```
user@k8s:$ kubectl exec <pod-name> -- ls /app
```

```
# run `ls /app` inside the pod
```



kubectl scale

adjust the number of
replicas for a deployment

```
user@k8s:$ kubectl scale deployment <deployment-name> --replicas=3  
# scale to 3 replicas
```



kubectl expose

create a service to expose
a pod or deployment

```
user@k8s:$ kubectl expose deployment <deployment-name> --  
type=NodePort --port=8080  
# expose a deployment on port 8080
```



kubecttl port-forward

forward local ports to a pod
for testing or debugging

```
user@k8s:$ kubectl port-forward <pod-name> 8080:80  
# forward local port 8080 to pod's port 80
```



kubectl roll-out

manage and monitor
deployment rollouts
(status, undo)

```
user@k8s:$ kubectl rollout status deployment/<deployment-name>  
# check the status of a rollout  
user@k8s:$ kubectl rollout undo deployment/<deployment-name>  
# rollback to the previous version
```



kubectl config

manage kubeconfig file
(for multiple cluster switching)

```
user@k8s:$ kubectl config get-contexts
```

```
# list available contexts.
```

```
user@k8s:$ kubectl config use-context <context-name>
```

```
# switch to a specific context.
```

```
user@k8s:$ kubectl config view
```

```
# view current configuration details.
```

